TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Carbon-Based Anodes for Low-Cost Anodic Electrochemical Production of Hydrogen Peroxide (H_2O_2)*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Chemical Processes	Process System Technologie	Greenhouse Gas Processing	Non-CO_2 Processing Technologies
Concept of Proposed Technology	Hydrogen peroxide (H_2O_2) is an essential chemical for sanitation, disinfection, and water treatment, yet the traditional anthraquinone (AO) process requires centralized large-scale plants, extensive solvent handling, and expensive transportation of concentrated peroxide. This creates high operating costs, safety risks, and supply limitations, particularly for hospitals, municipalities, and remote regions. We introduce a cost-efficient electrochemical solution capable of producing H_2O_2 directly where and when it is needed and using renewable electricity only. Our technology uses engineered carbon-based anodes to drive the two-electron water oxidation reaction ($2e^-$ WOR), eliminating the need for rare metals or expensive diamond-based electrodes. The result is a scalable, safe, and commercially attractive system for on-demand H_2O_2 generation powered only by water, electricity, and air, perfect for SMEs seeking opportunities in environmental,			

	healthcare, green-energy markets and sustainable chemistry.
Rationale for Joint R&D Project	This joint R&D aligns Fraunhofer IGB's expertise in sustainable electrochemical processes with the manufacturing strengths of Korean SMEs: - SMEs in Korea excel in carbon materials, electrochemical components, and device engineering, key elements for mass-produced H_2O_2 generators. - The market demand for compact, decentralized disinfection units is growing across hospitals, industrial cleaning, food processing, agriculture, and municipal water systems. - The technical approach of electrochemical synthesis of H_2O_2 meets the increasing need for "green" oxidants that can be generated on-demand and on-site, without using harmful raw materials and leaving no hazardous decomposition products. - This technology creates a new product category: portable or integrated electrolyzer units capable of generating H_2O_2 from water and electricity, lowering logistics costs and improving supply security. - Korean SME partners can commercialize devices across Asian and global markets, supported by Fraunhofer's technical validation and scale-up knowledge.
Key Words	1. Decentralized H_2O_2 production 2. Carbon electrodes 3. Flow electrolyzer 4. Green oxidants for disinfection & sanitation 5. Low-cost electrochemical technology

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level TRL 3	End of the Project	Level TRL 6
R&D Objectives	- Develop **low-cost, high-performance carbon electrodes** as a market alternative to expensive electrodes. - Engineer **flow-electrolyzer operation** to achieve multi-percent H_2O_2 concentrations with minimized decomposition. - Achieve **competitive H_2O_2 concentrations** (≥3 wt%) in continuous flow, enabling real-world sanitation/disinfection use cases. - Deliver a **prototype module ready for SME-led manufacturing** and product integration. - Build a technology foundation for **commercial H_2O_2 generators** targeting hospitals, water treatment operators, and industrial cleaning sectors.			
R&D Target Specifications	- Selectivity: FE(H_2O_2) ≥ 90% at relevant operating points. - Product concentration (continuous flow): ≥ 3 wt% H_2O_2. - Current density: ≥ 100 mA cm^{-2} (stable). - Operational stability: ≥ 500 h continuous run with minimal decay. - Electrode cost: ≤ 1,000 USD m^{-2} (materials + fabrication, excluding BoP). - Safety/quality: Suppressed H_2O_2 decomposition; product purity suitable for sanitation use			
R&D Plan	**Project Duration: 36 months total** • **Phase I (Pre-study): 6 months** • **Phase II (Main R&D): 30 months** **Phase I – Pre-Study (Months 1-6)** **Goal**: Validate low-cost carbon materials and key operating parameters - Electrode material screening - Binder & thermal treatment optimization - Initial flow-cell operation and performance benchmarking **Output**: Feasible carbon electrode with stable initial operation (>10 h) **Phase II – Main R&D (Months 7-36)** **WP2 (M7–M18): Electrode Engineering** - Surface modification, durability improvement - Scalability assessment for industrial fabrication - Milestone: Achieve ≥1 wt% H_2O_2 continuously **WP3 (M12–M30): Flow Reactor Optimization** - Process refinement for concentration, stability, and efficiency - Reliability testing and cost analysis - **Milestone**: Achieve 2–3 wt% H_2O_2 in stable flow operation **WP4 (M24–M36): Prototype Development** - Pilot-scale electrode production (roll-to-roll or lamination) - Assembly of a market-ready prototype "H_2O_2 Generator Module" - Field-relevant validation tests - **Milestone**: Commercializable prototype (≥3 wt%, >500 h operation)			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Fraunhofer IGB** - **Electrocatalyst & electrode development**: material screening, surface modification, binder/thermal optimization. - **Electrochemical testing**: selectivity, kinetics, degradation analysis, durability protocols. - **Flow-cell & process engineering**: cell design, hydrodynamics, suppression of side reactions, safety concepts. - **System integration support**: specifications for modules, quality control of electrodes, test methodologies
Expected R&R of Korean SME Partner	**Korean SME Partner** - Scale up electrode production using commercial manufacturing methods - System engineering of integrated prototype - Develop device housing, electronics, safety features, and modular system design - Conduct market-driven testing in sanitation/disinfection applications - Prepare commercialization strategy (distribution channels, certifications)

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	- **Technological**: Demonstration of carbon-based anodes enabling selective $2e^-$ WOR to H_2O_2 in continuous flow, meeting target specs (selectivity, concentration, durability, and cost). - **Industrialization pathway**: Prototype-ready electrode/device architecture compatible with SME-scale manufacturing and modular deployment. - **Societal & environmental**: Decentralized, on-demand H_2O_2 for disinfection and water treatment reduces transport of hazardous concentrations and supports resilient sanitation (Advanced Bio relevance). - **Strategic**: Reduced reliance on rare substrates; strengthens Korea-Germany collaboration in green electrochemical processing

6. References

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(2) Campos-Martín, J.M., **et al.** *Angew. Chem. Int. Ed.* **2006**, 45, 6962–6984 — Beyond the anthraquinone process

(3) Song, Y., **et al.** *J. Catal.* **2018**, 359, 68–75 — Electrocatalysis/thermal catalysis on carbon-supported metals.

(4) Park, S.Y., **et al.** *ACS Energy Lett.* **2019**, 4, 352–357 — $CaSnO_3$ for $2e^-$ WOR to H_2O_2.

(5) Kelly, S.R., **et al.** *ACS Catal.* **2019**, 9, 4593–4599 — ZnO as active/selective $2e^-$ WOR catalyst.

(6) Shi, X., **et al.** *Nat. Commun.* **2017**, 8, 701 — Activity trends in WOR to H_2O_2

(7) Pangotra, D., **et al.** *Green Chem.* **2023**, 24, 7931–7940 — **Anodic H_2O_2 in continuous flow**.

(8) Wenderich, K., **et al.** *ACS Sustain. Chem. Eng.* **2021**, 9, 7803–7812 — BDD evaluation.

(9) CO2EXIDE (EU H2020): CO_2-based electrosynthesis of ethylene oxide (https://co2exide.eu/).

(10) Pangotra, D., **et al.** *Appl. Catal. B* **2022**, 303, 120848 — **Anodic H_2O_2 using commercial carbons**.

(11) Power2Hype (EU Project): Electrochemical synthesis of hydrogen peroxide from water, air and renewable electric energy (www.power2hype.eu).

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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